

User Engagement & Health Outcomes Analysis

Evaluate user activation, retention, engagement patterns,
and associated health outcomes.

App Events
(User Behavior)

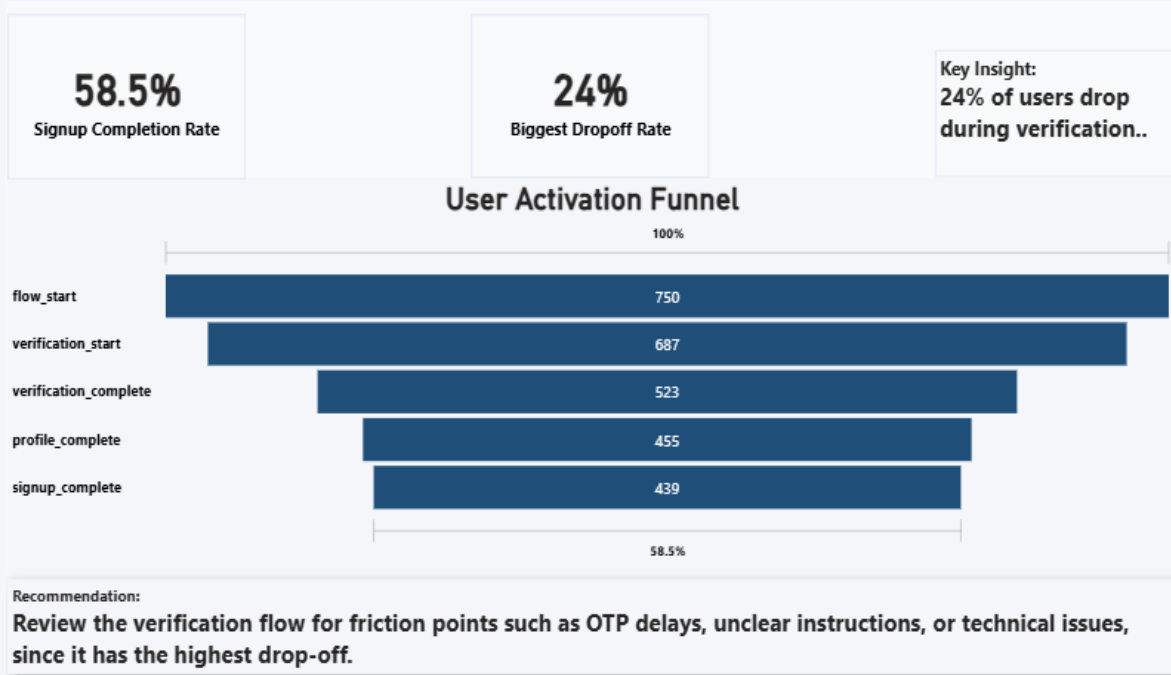
User Profiles
(Demographics)

Health Outcomes
(Weight & Body)

Tools: Python | SQL | Power BI

User Activation Funnel

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Key Insights

- 750 users initiated signup
- 439 users completed (~59%)
- Largest drop-off (~24%) at verification step

439 (58%)

Completion Rate: 59% | Biggest Drop-off: Verification Step (~24%)

Retention & Early Engagement

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52%

Average Week 1 Retention

Key Insight:

Week-1 retention averages 52%, but several cohorts drop below 40–45%, indicating inconsistent early engagement.

Retention Gap (High vs Moderate)

30%

Key Insight:

Early engagement strongly predicts retention (99% vs 69%)

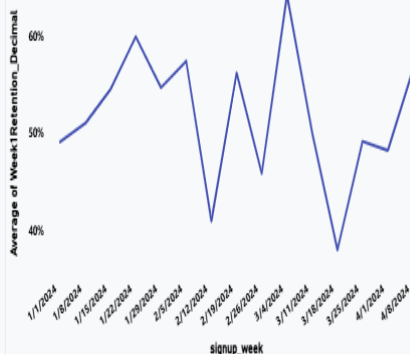
~52%

Avg Week 1 Retention

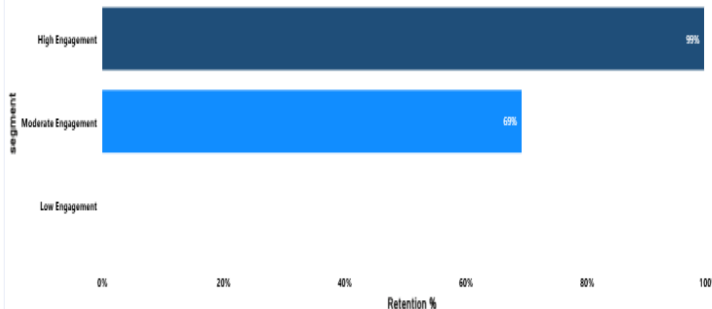
User Retention by Signup Cohort (Weekly)

signup week	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1/1/2024	55	27	30	25	29	28	27	29	29	26	28	30	29	28	28	28	15
1/8/2024	49	25	27	24	24	26	25	25	26	27	26	28	22	26	25	24	
1/15/2024	59	30	31	27	29	29	30	27	29	28	29	29	28	27	26		
1/22/2024	50	30	29	28	29	30	27	29	28	29	29	29	27	29			
1/29/2024	53	29	27	29	32	33	31	32	33	29	29	32	27				
2/5/2024	47	27	26	25	25	28	28	31	30	25	28	24					
2/12/2024	44	18	21	21	20	22	20	19	18	22	21						
2/19/2024	48	27	30	27	28	28	29	29	27	27							
2/26/2024	46	33	34	31	32	33	32	34	32								
3/4/2024	42	27	25	27	26	26	27	23									
3/11/2024	38	19	20	20	18	19	18										
3/18/2024	29	11	14	13	14	13											
3/25/2024	59	29	32	30	32												
4/1/2024	56	27	30	29													
4/8/2024	53	30	29	3													

Week 1 Retention Trend by Signup Cohort



Retention Rate by First-14-Day Engagement



Recommendation:

Encourage early activity during the first 14 days (e.g., onboarding prompts, reminders, or guided actions) to improve long-term retention.

~30%

Retention Gap

Recommendation:

Analyze cohorts with lower retention (<45%) to identify onboarding or engagement issues affecting early user retention.

Nearly half of new users do not return within the first week, early engagement is the single strongest predictor of long-term user retention.

Health Outcomes & Behaviors Pattern

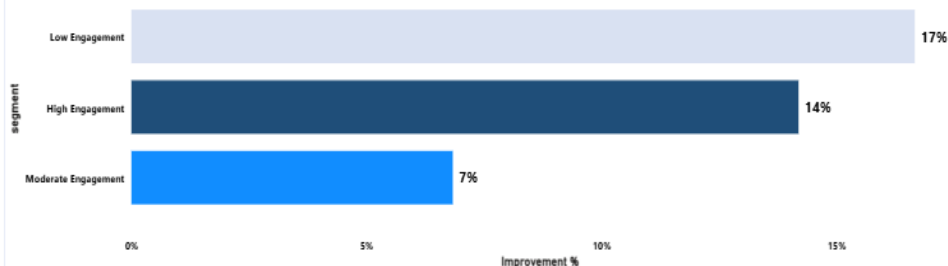
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Weight Improvement Gap

10%

Key Insight
Engagement
improves retention
but has a weaker link
to weight outcomes.

Weight Improvement by Engagement Level



Recommendation:

Combine engagement strategies with targeted health interventions to improve weight outcomes.

Weight improvement observed across ALL engagement segments

~10%

Weight Improvement
Gap

Key Points

1

Weight and body composition were tracked for up to 12 weeks where data was available.

2

A 5% weight reduction was used as the threshold for meaningful improvement.

3

Body fat data is only available for users with connected devices, so those findings represent a smaller group of users.

Improvement patterns vary, but consistent data logging appears linked to better outcomes.

Each message assessed for accuracy, clarity, tone, and safety. Ratings: Pass | Needs Edit | Fail

Issue & Rating	
A Needs Edit	Positive but slightly misleading — "keep up whatever you're doing" is vague and could imply medical advice.
B Fail	A 0.4% body fat change is very small and may reflect normal variation. Suggesting a metabolic issue could unnecessarily worry users.
C Needs Edit	Comparing the user to "most users" may feel discouraging and does not provide helpful context.
D Pass	Message clearly explains the improvement, links it to specific behaviours (steps, sleep). Tone is encouraging and informative.
E Needs Edit	Message may feel slightly negative or discouraging when no data has been recorded.

Improved Version
Your blood pressure decreased from 145/92 to 138/88 this week. Continuing healthy habits and following guidance from your healthcare provider may help support further improvement.
Your body fat increased slightly. Small changes can occur due to normal variation — it may be helpful to continue tracking over time.
You logged 4 meals this week. Logging more consistently can help us provide more accurate insights about your nutrition patterns.
<i>No changes needed — message is clear, positive, and supported by data.</i>
We didn't see any new data recorded this week. Logging your activity tomorrow can help keep your progress up to date.

AI health messages should balance encouragement with caution — avoid implying medical guidance from small or ambiguous data changes.

Recommendations & Next Steps

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Product Recommendations

01

Streamline Verification

Simplify verification to reduce onboarding drop-off and improve activation rates.

02

Drive Early Engagement

Encourage engagement in the first 14 days via reminders or guided actions.

03

Promote Device & Logging

Push device connections and health logging at onboarding to improve data completeness.

04

Refine AI Insights

Continuously refine AI-generated insights to keep them accurate, supportive, and safe.

With More Time or Data

1

Segmented Engagement Analysis

Explore how engagement patterns vary across markets, subscription tiers, or referral sources to identify highest-value user segments.

2

Long-Term Retention & Outcomes

A longer observation period would allow deeper analysis of retention curves beyond Week 1 and more robust health outcome measurement.

3

Expanded Behavioural Signals

Education content usage, meal logging patterns, and coach interactions could reveal what truly drives long-term user success.

Acting on these recommendations now, while building toward deeper analysis, will drive sustainable improvements in user outcomes.